

A03 Neural Network Zoo Contribution Journal

The CNN Eagle

Team: DataProxy

Course: ITAI 2376

Team Members: Elijah Raines, Justin Davis, Samantha Mireles, Osemudia Ogedengbe, Brandon Matias, and Jonah Joseph

Project Contribution Overview

Our group divided the Neural Network Zoo assignment into connected sections so each member could explain one part of the CNN Eagle project. The presentation began with the basic ideas behind neural networks and deep learning, moved through the structure and learning process of a CNN, and ended with applications, a demonstration, and a group reflection. Each contribution supported the same goal: explaining a technical topic in a clear way that connected CNN image recognition to the sharp vision of an eagle.

Team Roles and Contributions

Team Member	Primary Role	Contribution to A03
Elijah Raines	CNN layer structure	Explained the input, hidden, and output layers. His section showed how an image enters the model as pixel data, how hidden layers find visual features, and how the output layer produces the final prediction.
Justin Davis	Neural network and deep learning foundation	Helped introduce neural networks as connected layers that learn patterns from examples. He also explained how deep learning uses multiple layers to build more detailed features before making a prediction.
Samantha Mireles	CNN selection and eagle comparison	Explained why the group selected a Convolutional Neural Network and why an eagle was a useful visual example. Her section connected an eagle's sharp vision with the way CNNs find edges, shapes, colors, textures, and patterns.
Osemudia Ogedengbe	CNN mechanics and learning process	Covered convolution, filters or kernels, feature maps, pooling, flattening, and backpropagation. This section explained how a CNN finds useful image features and improves by adjusting after prediction errors.
Brandon Matias	Applications and interactive demonstration	Developed the real-world examples for medical imaging, autonomous driving, and facial recognition. He also organized the interactive model demonstration that showed how inputs move through hidden layers to class probabilities.
Jonah Joseph	Reflection and closing takeaway	Prepared the final reflection about choosing the right neural network for the data and task. His conclusion explained that different network architectures specialize in different problems, just as animals have different strengths.

How the Contributions Worked Together

The sections were designed to build on one another. Justin's introduction established the basic idea of layered learning. Samantha's CNN and eagle comparison gave the class a simple visual way to understand image recognition, while Elijah's layer breakdown showed how information moves through the network. Osemudia then explained the steps that make the CNN work and learn. Brandon connected those ideas to technology people see in real life and to the interactive demo. Jonah closed the presentation by explaining why the best neural network depends on the problem being solved.

This order helped the presentation move from simple definitions to a complete example. The early sections gave the audience the vocabulary needed for the later sections, while the applications and reflection showed why the technical details matter outside the classroom.

Teamwork Takeaways

Dividing the assignment by topic gave each member a clear responsibility. The group could cover more information without asking one person to explain the entire CNN process. Each role also depended on the others. The eagle comparison made the technical sections easier to picture, and the technical sections gave the real-world examples a stronger explanation.

The final presentation shows that a group project works best when each contribution supports the same main idea. The terminology, examples, demo, and reflection all focused on how a CNN recognizes visual patterns and improves through training. That common focus made the individual sections feel like one project instead of separate reports.

Contribution Summary

Every member contributed a necessary part of A03. Elijah and Justin established the neural network foundation, Samantha developed the CNN Eagle comparison, Osemudia explained the internal CNN process, Brandon connected the lesson to applications and the interactive model, and Jonah summarized the main lesson about choosing the right network. Together, the group covered both how a CNN works and why it is useful.